ROHAN CHAUHAN

Python AI / LLM / Backend and Web Application Developer

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WHO I AM

Software Engineer specializing in AI-driven backend systems, with hands-on experience shipping production LLM, RAG and agentic applications, OCR-based document pipelines and event-driven services. Skilled in Python, FastAPI, Django, Kafka, Celery and Redis, with data work across PostgreSQL, MongoDB and BigQuery and deployments on Docker, AWS and Vertex AI. Focused on building scalable, reliable systems that automate real business workflows.

WORK EXPERIENCE

Moglix – B2B Industrial Commerce & Supply Chain

Software Engineer — July 2026 - Present

- Built an AI cash-application pipeline (LayoutLMv3, Qdrant, Gemini) that auto-maps customer payments to invoices.
- Engineered a GenAI invoice pipeline (Gemini, AWS Textract) streaming data via Kafka → MongoDB → BigQuery.
- Built a self-learning parser loop where HDBSCAN clusters new document layouts and Gemini writes parsing configs.
- Develop LLM agents and REST APIs with the AI & Analytics team, including a WhatsApp procurement assistant.
- **Tools & Tech:** Python, FastAPI, Django, Celery, Kafka, Redis, MongoDB, BigQuery, Gemini, Vertex AI, OCR, Docker

Python Engineering - Intern — Mar 2026 - July 2026

- Built Python ETL pipelines with Pandas and SQL across MySQL, PostgreSQL and MongoDB for analysis-ready data.
- Developed REST APIs and Kafka consumers to automate recurring workflows such as Kaleyra SMS notifications.
- Partnered with analytics teams to replace manual reporting with automated, data-driven jobs.
- **Tools & Tech:** Python, Pandas, REST APIs, Kafka, MySQL, PostgreSQL, MongoDB, Kaleyra, Git

Coding Jr – EdTech Platform

AI-Research - Intern — June 2025 - Jan 2026

- Built backend logic, REST APIs and core features in Python for the company's internal admin panel as an SDE.
- Researched and prototyped NLP and LLM-based features to automate manual operations on the platform.
- **Tools & Tech:** Python, REST APIs, NLP, Web Design, Git

EDUCATION

Schooling / Degree	School / University	CGPA / Percentage	Duration
B.Tech CSE (AI & ML)	Greater Noida Institute of Tech.	8.3 CGPA	2022 - 2026
Intermediate (PCM)	St Mary's School Kashipur	78%	2021
High School	St Mary's School Kashipur	78%	2019

PUBLICATIONS & CERTIFICATIONS

BLOCKAUTH - A BLOCKCHAIN-BASED DECENTRALIZED IDENTITY MANAGEMENT SYSTEM

Research Paper Presentation | ICSDS-2025 | Hooghly Engineering & Technology College | Dec 2025

- Designed and developed a blockchain-based decentralized identity management system for secure authentication and access control across IoT networks.
- Presented the research at ICSDS-2025, demonstrating how decentralized identity enables tamper-resistant, privacy-preserving access control.

Machine Learning A-Z: AI, Python & R — Udemy | 2023

Intel Unnati Program – AI/ML Track — Intel | 2025

Introduction to Cybersecurity — Cisco | 2025

AWS Deep Dive — AWS | 2023

Deloitte Data Analytics Job Simulation — Forage | 2025

AWS Cloud Projects — Coursera | 2024

SKILLS

Languages:	Python, TypeScript, SQL, JavaScript
AI / ML:	PyTorch, TensorFlow, Scikit-learn, XGBoost, NLP, OCR, Optuna, Hugging Face
GenAI & LLMs:	RAG, Agentic AI, LangChain, LangGraph, LoRA, LLM Fine-tuning, vLLM, Ollama, Gemini API, Vertex AI
Backend & Systems:	FastAPI, Django, REST APIs, Microservices, Apache Kafka, Celery, Redis
Databases:	PostgreSQL, MongoDB, BigQuery, FAISS, ChromaDB, Qdrant
Data & DevOps:	ETL, Pandas, NumPy, AWS, Docker, Git, CI/CD

PROJECTS

Ramanujan: Agentic Conversational Analytics Assistant
An AI analytics copilot that lets business teams ask questions in plain English inside Microsoft Teams and get answers from enterprise data in seconds, with no SQL and no dashboards.
Tech used: Python, FastAPI, Gemini, Vertex AI, BigQuery, Qdrant, Sentence-Transformers, Microsoft Graph API, Docker

- Built a Gemini ReAct agent that turns natural-language questions into governed BigQuery SQL. It uses RAG over a Qdrant-backed KPI glossary with sentence-transformer embeddings, so answers match the business definition of each metric.
- Engineered a production-grade FastAPI service on Vertex AI using hexagonal architecture. Safety layers include a SQL policy engine with row-level access control and grounding guardrails against hallucinated answers.

Lumida: AI Data Anomaly Detection Engine
A self-hosted engine that connects read-only to company databases, finds hidden data problems, explains them with an LLM, and learns from human feedback.
Links: lumida.netlify.app | github.com/RohanCh07/Lumida
Tech used: Python, FastAPI, PostgreSQL, SQLAlchemy, Docker, Google Gemini, REST APIs.

- Engineered a data-quality engine for 5 databases (PostgreSQL, MySQL, SQL Server, Oracle, Snowflake) that infers undeclared table relationships and detects missing data, volume drops, outliers, stale tables and misplaced personal data.
- Built a human-in-the-loop AI workflow where Gemini proposes business rules that, once approved, run nightly as SQL checks. It found 15/15 hidden relationships and 13/13 planted issues with 0 false alarms, backed by 353 automated tests.

Payment Advice Automation Platform: AI-Driven Cash Application for Accounts Receivable
An AI-driven pipeline at Moglix that reads customer payment advices from Freshservice tickets, extracts UTRs, invoices and TDS, validates them against ERP, and auto-maps receipts to open invoices for the finance AR team.
Tech used: Python, Django, Celery, Kafka, Redis, MongoDB, MySQL, Qdrant, LayoutLMv3, HDBSCAN, Gemini (Vertex AI), PyMuPDF, HTMX, Docker

- Engineered an end-to-end cash-application pipeline: tickets flow from Freshservice through Kafka and Celery, and LayoutLMv3 classification with Qdrant layout matching routes each advice to one of 14+ customer-specific parsers for large enterprise clients, with Gemini as fallback. Extracted payments are validated against ERP, matched to bank entries, and auto-mapped behind open-balance and deduction guards. 100+ page advices are parsed in about 0.2 s.
- Built a self-learning onboarding loop in which HDBSCAN clusters unseen layouts and Gemini writes per-cluster regex configs, auto-activated above 80% held-out accuracy. Delivered a Django/HTMX dashboard for per-ticket tracing and bulk reprocessing, and led the Gemini 2.5 Flash → 3.5 Flash Lite migration with an A/B harness for about 3× faster, cheaper extraction.

SupplierInv: AI-Powered Invoice Intelligence Pipeline
An automated invoice processing system built at Moglix that extracts, validates, and reconciles supplier invoices at scale using generative AI.
Tech used: Python, Gemini, AWS Textract, Pydantic, Kafka, MongoDB, BigQuery, Power BI

- Built a GenAI document pipeline using Gemini and AWS Textract that reads unstructured invoices and turns them into schema-validated data. A real-time Kafka → MongoDB → BigQuery backbone streams the data at enterprise scale.
- Automated reconciliation with fuzzy item matching, tolerance bands, HSN validation, and GSTIN compliance checks, replacing hours of manual review with zero-touch processing. The results feed live Power BI dashboards for procurement analytics.

Conversational Procurement Intelligence Platform
An AI-powered procurement assistant accessible through WhatsApp, enabling users to interact with sourcing workflows through natural-language conversations.
Tech used: Python, FastAPI, WhatsApp Business API, Agentic AI, REST APIs

- Engineered a WhatsApp-integrated AI agent to facilitate conversational procurement workflows, enabling users to interact with sourcing operations through natural-language queries.
- Developed backend services and agent orchestration pipelines to process user requests, execute sourcing workflows, and deliver contextual responses through WhatsApp.

Distiller: Self-Hosted LLM Cost Optimizer
An OpenAI-compatible gateway that learns a company's repeated LLM tasks and quietly moves them to a small, fine-tuned model it owns, cutting API costs without changing application code.
Links: github.com/RohanCh07/distiller (in development)
Tech used: Python, FastAPI, LiteLLM, LoRA, Ollama, PostgreSQL, Docker, CLI Tooling

- Engineered a drop-in OpenAI-compatible gateway that teams adopt by changing a single base_url. It logs every prompt and response so repeated task patterns can be identified automatically.
- Architected a distillation pipeline that fine-tunes small open-source models with LoRA on the logged outputs of large models. Each model is validated in shadow mode before routing, and calls fall back to the large model whenever the small one is unsure.

Repo-Safety: "Is This Repo Safe?" Malware Scanner
A static security scanner that checks untrusted repos and ZIPs, like recruiter "take-home tests", for hidden auto-run malware before a developer opens or runs them.
Links: github.com/RohanCh07/Repo-Safety
Tech used: TypeScript, Node.js, Fastify, BullMQ, Redis, PostgreSQL, Docker, VS Code Extension API, Chrome Extension (MV3), Vitest

- Engineered a hostile-input scanning engine with 11 detection rules (VS Code folder-open tasks, install hooks, hidden git hooks, download-and-run scripts, obfuscated code, typosquatted dependencies). It returns a Safe / Suspicious / Dangerous verdict with file-and-line evidence, and never executes code or runs git clone.
- Architected one engine behind four surfaces (CLI, VS Code extension, web app and Chrome extension), hardened against zip bombs, zip-slip and SSRF. It scanned the 68k-file TypeScript repo in 12 seconds and flagged none of 21 popular open-source repos as Dangerous.